

GITHUB WEEK - 3 ASSIGNMENT

Python Program to Print the Fibonacci Series

Problem Statement

Write a Python program to print the **Fibonacci series for n terms**.

1. Objective

To write and execute a Python program that prints the **Fibonacci series for a given number of terms** using a loop.

2. Problem Description

The Fibonacci series is a sequence of numbers in which each number is obtained by adding the previous two numbers. The series starts with **0, 1**.

For example: $0 + 1 = 1$, $1 + 1 = 2$, $1 + 2 = 3$, $2 + 3 = 5$

Therefore, the Fibonacci series is: **0 1 1 2 3 5 8**

When the input is 7, the expected output is:

```
Fibonacci Series:  
0  1  1  2  3  5  8
```

3. Algorithm

1. Start the program.
2. Get the number of terms from the user.
3. Initialize the first Fibonacci number as 0.
4. Initialize the second Fibonacci number as 1.
5. Use a `for` loop for the given number of terms.
6. Display the current Fibonacci number.
7. Calculate the next Fibonacci number by adding the previous two numbers.
8. Update the first and second Fibonacci numbers.
9. Stop the program.

4. Python Program

File Name: `fibonacci.py`

```
# Program to print the Fibonacci series for n terms  
  
n = int(input("Enter the number of terms: "))
```

```

a = 0
b = 1

print("Fibonacci Series:")

for i in range(n):
    print(a, end=" ")    a, b
    = b, a + b

```

5. Program Explanation

The program first gets the number of terms from the user using the `input()` function.

```
n = int(input("Enter the number of terms: "))
```

The first two Fibonacci values are initialized as 0 and 1.

```

a = 0
b = 1

```

The program displays the heading using the `print()` function.

```
print("Fibonacci Series:")
```

A `for` loop is then used to print the series for the required number of terms.

```

for i in range(n):
    print(a, end=" ")
    a, b = b, a + b

```

Here, `a` represents the current Fibonacci number. The statement `a, b = b, a + b` updates the values so that the next Fibonacci number can be generated.

For example:

```

Start:                a = 0, b = 1
After first update:   a = 1, b = 1
After second update:  a = 1, b = 2

```

This process continues until the required number of terms is printed.

6. Sample Input

```
Enter the number of terms: 7
```

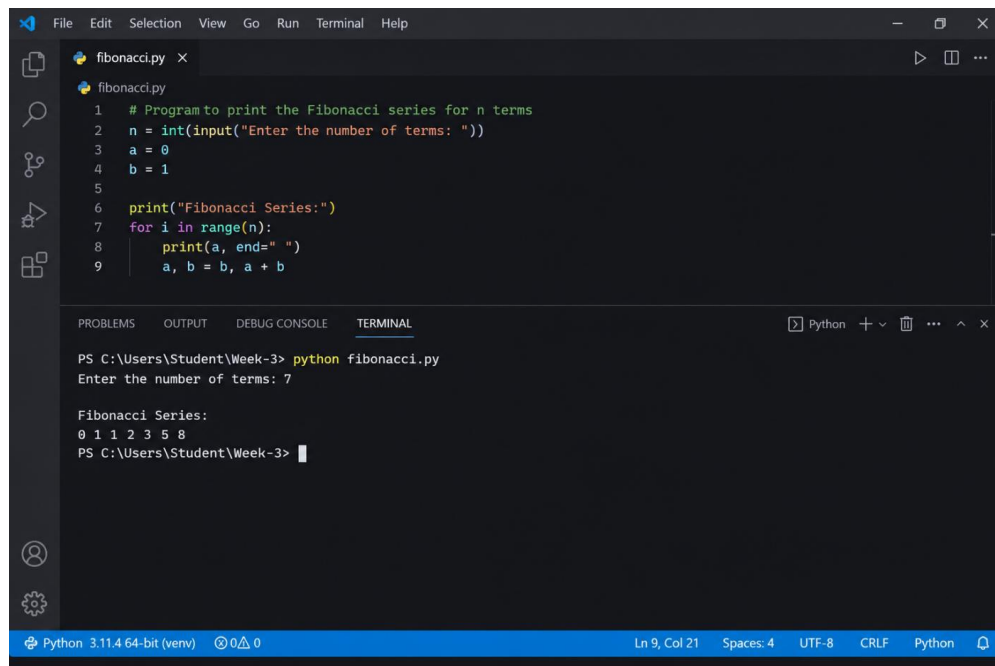
7. Sample Output

```

Fibonacci Series:
0  1 1 2 3 5  8

```

8. Output Screenshot *Figure 1: Output of Fibonacci Python Program*



The screenshot shows a Visual Studio Code editor window with a file named `fibonacci.py`. The code is as follows:

```
1 # Program to print the Fibonacci series for n terms
2 n = int(input("Enter the number of terms: "))
3 a = 0
4 b = 1
5
6 print("Fibonacci Series:")
7 for i in range(n):
8     print(a, end=" ")
9     a, b = b, a + b
```

The terminal window at the bottom shows the execution of the program:

```
PS C:\Users\Student\Week-3> python fibonacci.py
Enter the number of terms: 7

Fibonacci Series:
0 1 1 2 3 5 8
PS C:\Users\Student\Week-3> |
```

9. GitHub Repository Structure

```
Week-3/
|-- fibonacci.py
|-- README.md
`-- output.png
```

`fibonacci.py` - Python source code

`README.md` - Project description and instructions

`output.png` - Screenshot of the program output

10. GitHub Submission Steps

Step 1 - Create the Python file: Create `fibonacci.py` and paste the program into it.

Step 2 - Execute the program: Run it and enter 7. The output is 0 1 1 2 3 5 8.

Step 3 - Take screenshot: Capture the full execution and save it as `output.png`.

Step 4 - Create README: Create `README.md` with the project description.

Step 5 - Upload to GitHub: Upload `fibonacci.py`, `README.md` and `output.png` to the Week-3 repository.

Git commands:

```
git add fibonacci.py README.md output.png git commit
-m "Added Week 3 Fibonacci series program" git push
```

11. Result

The Python program was successfully implemented and executed to print the Fibonacci series for a given number of terms. For the input **7**, the program produced **0 1 1 2 3 5 8**. The Python program, README file, and output screenshot were uploaded to the GitHub repository successfully.

12. GitHub Repository Link

GitHub Repository: [[Paste your GitHub repository link here](#)]

Example: <https://github.com/your-username/week-3>